



ORIGINAL ARTICLE

Determination of Mineral Nutrients and Toxic Elements in Brazilian Soluble Coffee by ICP-AES

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The determination of mineral nutrients and toxic elements Na, K, Mg, Al, P, S, Ca, Mn, Fe, Ni, Cu, Zn, Cd, Sb, Pb, Cr and Sn, was made in 21 samples of Brazilian soluble coffee by inductively coupled plasma atomic emission spectrometry (ICP-AES), after sample preparation in a focused microwave system with HNO₃ and H₂O₂ reagents. Matrix interference was detected in the determination of Sn and the correction of the background enhancement was made by the equipment software. The results obtained for mineral nutrients and toxic elements were acceptable to human consumption at nutritional and toxic levels except for a sample with 0.52 ± 0.02 mg/kg of Cr, a value above the specified one in the Brazilian Food Legislation. The application of the Principal Components Analysis and the Agglomerative Hierarchical Clustering on the results enabled the investigation of the variability and the classification of groups, resulting in three clusters with similarities in the mineral compositions.

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Key Words: soluble coffee; nutrients and toxic elements; ICP-AES.

INTRODUCTION

Coffee is one of the more important agricultural products in the international trade. Brazil continues to be the greatest producer and exporter of this product. Basically two types of coffee are produced in Brazil: *arabica coffea* and *coffea robust canephora*. The Brazilian production arrives from Minas Gerais, Espírito Santo, São Paulo and Paraná states and a small parcel is produced in Rondônia and Bahia states. Coffee, in terms of international trade, is the second product in value, putting into motion approximately US\$ 35 billion per year and being supplanted only by petroleum. The production of Brazilian soluble coffee is in the greatest part destined for the foreign market with the invoicing around US\$ 500 million (Silva, 1996). The determination of mineral nutrients and toxic elements is of great interest, considering the great consumption of this product for millions of people world-wide. Thus, the development and improvement of chemical analysis methods are very important for the research on these contaminants and nutrients. Studies on mineral nutrients and toxic elements in coffee (green coffee, roasted and soluble coffee) frequently use the technique of spectrometry of atomic absorption (AAS) and graphite furnace (GFAAS) with

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conventional digestion methods that include low-temperature ashing and wet digest method (Horwitz and Van Der Linden, 1974; Lara *et al.*, 1975/76; Krivan *et al.*, 1993; Koch *et al.*, 1989; Suzuki *et al.*, 1982; Gillies and Birkbeck, 1983; Kapur and West, 1974; Kuennen *et al.*, 1982). Few studies used ICP-AES (Koch *et al.*, 1989) and XRFS (Orlic *et al.*, 1986). An appropriate digestion procedure for soluble coffee, in order to determine the nutrients and toxic elements by ICP-AES, was established by the authors (Santos and Oliveira, 1997). A quick and safe digestion of the samples has been obtained through the focused microwave system. Therefore, the inductively coupled plasma atomic emission spectrometry (ICP-AES) is an excellent technique for the determination of major, minor, and trace elements in food samples because of its low detection limit, high sensitivity, wide linear dynamic range, relatively high freedom from spectral, chemical and ionization interferences, precision, speed and versatility in terms of the number of elements determined (Munilla *et al.*, 1995; Montaser and Golightly, 1987). So ICP-AES could offer considerable advantages for the quantitative analysis of food such as soluble coffee. The objective of this work was the determination of mineral nutrients and toxic elements in soluble coffee by ICP-AES, in order to verify the quality of the composition of Brazilian soluble coffee coming from a specific region of the country.

MATERIALS AND METHODS

Equipment and Reagents

Digestions were performed on a focused microwave system model 7400 Spex, using borosilicate vessels. All measurements were performed with a Spectroflame Modula optical atomic emission spectrometer from Spectro Co. The instrumental conditions are presented in Table 1.

The parameters of the analytical curve for all studied elements are presented in Table 2. Stock solutions (1000 µg/mL) from Spex and Titrisol Merck were used for the preparation of the standards in 1% (v/v) HNO₃. The selected wavelength showed no elevation of background, spectral and matrix interferences for most elements with the exception of Sn which showed an elevation of background that was corrected by the software from Spectroflame Modula ICP-AES (Fig. 1).

The detection limit was the instrumental limit, calculated as three times the standard deviation of the blank expressed in concentration and the practical detection limit was obtained according to Thompson (1987). Distilled and deionized water (Permutation water purification unit) was used to prepare samples and standards. All reagents used were of analytical grade.

TABLE 1
Instrumental conditions used in ICP-AES

Power	1.2 kW
Flow of Ar refrigerant	12 L/min
Flow of Ar auxiliary	1.2 L/min
Flow of Ar carrier	1.0 mL/min
Pressure in the Meinhard nebulizer	179.2 kPa
Introduction of the sample	1.5 mL/min
Observation height	12 mm

TABLE 2
Parameters of the analytical curve for studied elements¹

Element	λ (nm)	A_0	A_1	LD ($\mu\text{g/mL}$)	Practical LD ($\mu\text{g/mL}$)
Na	588.995	- 0.7743	1.0369×10^{-4}	0.02	0.1
K	766.491	- 9.7663	1.5488×10^{-3}	0.07	1.0
Mg	285.213	- 0.3579	9.1608×10^{-5}	0.01	0.05
Al	167.080	- 1.0510	1.6801×10^{-5}	0.02	0.1
P	178.290	- 0.3985	1.6906×10^{-4}	0.01	0.05
S	182.040	- 0.9916	1.5915×10^{-4}	0.02	0.1
Ca	422.673	- 0.5674	2.7911×10^{-5}	0.01	0.1
Mn	293.930	- 0.6073	2.9262×10^{-4}	0.01	0.05
Fe	259.940	- 1.0930	2.6105×10^{-3}	0.02	0.1
Sn	189.926	- 0.1410	9.3238×10^{-4}	0.01	0.1
Cr	357.869	- 0.3776	8.9826×10^{-5}	0.005	0.01
Ni	352.454	- 1.3386	4.1800×10^{-4}	0.03	0.05
Cu	324.754	- 0.08817	1.5887×10^{-5}	0.002	0.01
Zn	213.856	- 0.07228	5.9600×10^{-6}	0.001	0.005
Cd	226.502	- 0.06134	9.9533×10^{-6}	0.001	0.005
Sb	217.581	- 0.8281	1.7265×10^{-4}	0.01	0.04
Pb	220.353	- 2.2400	1.0236×10^{-3}	0.05	0.10

¹ A_0 = linear coefficient, A_1 = angular coefficient and LD = detection limit.

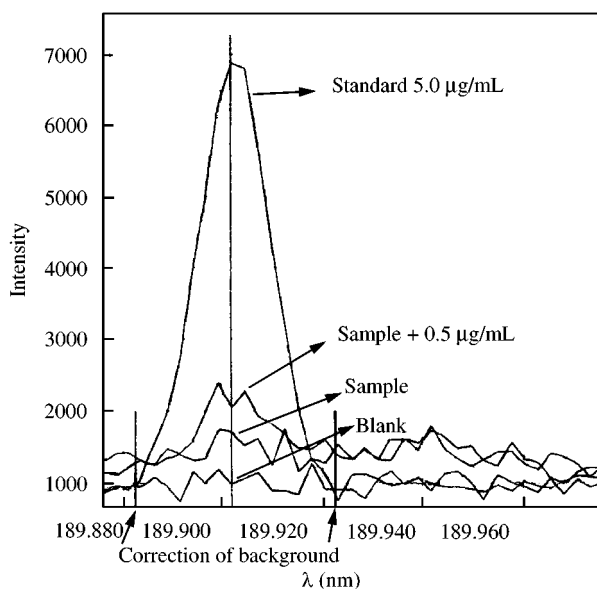


FIGURE 1. Elevation of Sn background.

Contamination Control

To avoid contamination from the recipients and volumetric flasks used, they were washed with detergent, soaked in a 1 mol/L HNO_3 solution, rinsed with deionized water, filled with 10 mol/L HNO_3 , kept for at least 1 night and again rinsed with deionized water. The digestion vessels were cleaned by adding 10 mL of 65% (v/v)

HNO₃ and submitting them to the same microwave program applied to the samples. Any contamination from the vessels was monitored by analyzing digestion blanks.

Samples

Twenty-one samples of soluble coffee (marks: Café Pelé, Gran Classic, Nescafé Matinal, Nescafé Tradição, Café do Ponto Freeze-dried, Café do Ponto Descafeinado, Café Solúvel Parmalat, Ariscafé, Café Solúvel Iguazu, Café Solúvel Eldoro, Café Solúvel Diana and Bassil Coffee) were collected in the Brazilian market and were used in the present work.

Analytical Procedures

Sample digestion for determination of Na, K, Ca, Mg, P, Fe, S, Mn, Cu, Zn, Sb, Sn, Ni, Cd and Al. 1 g of the sample was introduced into a 250 mL glass tube (from the focused microwave system model 7400 SPEX) and 6 mL of 65% (v/v) HNO₃ were added. The glass tube was placed on the focused microwave system, heated at 105 W power for 10 min, and allowed to cool for 5 min. Then 0.5 mL of 30% (v/v) H₂O₂ was added, and the heating and cooling procedures were repeated once more. The solutions were diluted up to 50 mL.

Sample digestion for determination of Pb and Cr. In order to comply with the Brazilian Food Legislation that establishes the maximum Pb content as 1.0 µg/g and Cr as 0.1 µg/g, 5 g of the sample was weighed. It was introduced into a 250 mL glass tube (from the focused microwave system model 7400 SPEX) and 30 mL of 65% (v/v) HNO₃ were added. After 30 min of initial reaction at ambient temperature, the glass tube was placed on the focused microwave system, heated at 105 W power for 5 min, and allowed to cool for 5 min. The heating and cooling procedures were repeated. Then 2.5 mL of 30% (v/v) H₂O₂ were added, and the heating and cooling procedures were repeated once more. The solutions were diluted up to 50 mL.

TABLE 3
Validation of methods using spike samples of soluble coffee¹

Element	Measured value	Added value	Recovery value (n = 3)	Recovery %
Mg	0.334 ± 0.004 g/100 g	0.250 g/100 g	0.206 ± 0.005 g/100 g	82.4 ± 2.2
Al	< 5.0 mg/kg	25.0 mg/kg	21.0 ± 1.0 mg/kg	84.0 ± 4.0
P	0.335 ± 0.010 g/100 g	0.250 g/100 g	0.236 ± 0.018 g/100 g	94.4 ± 7.2
Ca	0.131 ± 0.003 g/100 g	0.125 g/100 g	0.118 ± 0.012 g/100 g	94.4 ± 9.6
Mn	7.00 ± 0.50 mg/kg	25.0 mg/kg	24.5 ± 2.0 mg/kg	98.0 ± 8.0
Fe	38.0 ± 2.5 mg/kg	50.0 mg/kg	47.0 ± 3.5 mg/kg	94.0 ± 7.0
Sn	7.5 mg/kg	25.0 mg/kg	26.0 ± 0.3 mg/kg	104 ± 1
S	0.180 ± 0.008 g/100 g	0.125 g/100 g	0.139 ± 0.014 g/100 g	111 ± 11
Na	0.0580 ± 0.0015 g/100 g	0.0500 g/100 g	0.0495 ± 0.005 g/100 g	99.0 ± 10.0
K	3.77 ± 0.06 g/100 g	0.50 g/100 g	0.41 ± 0.02 g/100 g	82.5 ± 4.0
Cr	< 0.10 mg/kg	5.0 mg/kg	4.8 ± 0.3 mg/kg	96.0 ± 6.0
Ni	< 2.50 mg/kg	25.0 mg/kg	20.0 ± 0.5 mg/kg	80.0 ± 2.0
Cu	1.55 ± 0.15 mg/kg	25.0 mg/kg	23.5 ± 0.5 mg/kg	94.0 ± 2.0
Zn	3.75 ± 0.35 mg/kg	25.0 mg/kg	21.5 ± 1.5 mg/kg	86.0 ± 6.0
Cd	< 0.25 mg/kg	25.0 mg/kg	22.0 ± 0.5 mg/kg	88.0 ± 2.0
Sb	< 2.00 mg/kg	25.0 mg/kg	21.5 ± 1.0 mg/kg	86.0 ± 4.0
Pb	< 1.00 mg/kg	5.0 mg/kg	5.1 ± 0.5 mg/kg	102 ± 10

¹ < not detectable at level.

Validation of Methods

The accuracy and validity of the measurements were determined by analyzing a spike sample of soluble coffee. Three replicate analyses were carried out and these spiked samples were treated as described in analytical procedures. The results obtained are presented in Table 3, showing a good recovery for all the elements studied.

RESULTS AND DISCUSSION

The results obtained for the macro-, micro-minerals and toxic elements expressed as the mean and standard deviation (S.D.) of two replicates analyses for the 21 samples of Brazilian soluble coffee are presented in Tables 4 and 5.

The toxic elements Ni, Cd, Sb and Pb were not detected at levels of 2.50; 0.25; 2.0 and 1.0 mg/kg, respectively.

According to Debry (1994) the mineral content of soluble coffee is on average: Na 0.05 g/100g; K 4 g/100 g; Ca 0.15 g/100 g; P 0.35 g/100 g; Fe 45 mg/kg; Cu 0.5 mg/kg and Zn 5 mg/kg. The S content in green coffee is on average 0.13 g/100 g and the Mg content varies from 0.16 to 0.31 g/100 g. Suzuki *et al.* (1982) reported a concentration of Cu of 2.4 mg/kg. Lara *et al.* (1975/76) worked with coffee drink, green and roasted coffee, and reported an average concentration of 1.2 mg/kg for Cu. The obtained results are on the same level as these previous ones. Fe and Na were present in a wide range of concentration among the samples. This variation could be attributed to the type of soil where the coffee was cultivated and/or to the differences in the industrial processes producing soluble coffee. The Brazilian Food Legislation (Compêndio da Legislação de Alimentos, 1988) has established maximum values only for Cu (30 mg/kg) and Zn (50 mg/kg), and none of the samples analyzed presented values close to these.

TABLE 4
Macro-minerals (g/100 g) present in Brazilian soluble coffee

Samples	Na	K	Mg	P	S	Ca
01	0.0583 ± 0.0012	3.63 ± 0.08	0.327 ± 0.007	0.336 ± 0.005	0.173 ± 0.004	0.135 ± 0.003
02	0.0531 ± 0.0008	3.46 ± 0.02	0.311 ± 0.006	0.316 ± 0.004	0.170 ± 0.002	0.130 ± 0.001
03	0.1550 ± 0.0063	3.47 ± 0.14	0.312 ± 0.015	0.358 ± 0.012	0.172 ± 0.008	0.150 ± 0.006
04	0.1743 ± 0.0042	3.65 ± 0.04	0.384 ± 0.014	0.361 ± 0.007	0.199 ± 0.002	0.159 ± 0.002
05	0.1483 ± 0.0067	3.92 ± 0.16	0.392 ± 0.018	0.372 ± 0.006	0.191 ± 0.004	0.167 ± 0.003
06	0.1084 ± 0.0025	4.58 ± 0.08	0.415 ± 0.024	0.400 ± 0.009	0.206 ± 0.005	0.146 ± 0.004
07	0.0322 ± 0.0019	4.73 ± 0.16	0.287 ± 0.02	0.272 ± 0.006	0.148 ± 0.004	0.136 ± 0.005
08	0.0325 ± 0.0022	3.89 ± 0.21	0.360 ± 0.023	0.344 ± 0.020	0.184 ± 0.012	0.125 ± 0.006
09	0.0363 ± 0.0007	5.17 ± 0.20	0.300 ± 0.019	0.297 ± 0.007	0.175 ± 0.005	0.142 ± 0.004
10	0.0274 ± 0.0002	3.46 ± 0.11	0.372 ± 0.005	0.410 ± 0.013	0.182 ± 0.004	0.156 ± 0.002
11	0.0322 ± 0.0005	3.58 ± 0.04	0.374 ± 0.003	0.344 ± 0.003	0.156 ± 0.001	0.116 ± 0.001
12	0.0290 ± 0.0005	3.48 ± 0.04	0.329 ± 0.005	0.356 ± 0.006	0.169 ± 0.001	0.111 ± 0.001
13	0.3225 ± 0.0018	3.98 ± 0.02	0.212 ± 0.006	0.223 ± 0.002	0.175 ± 0.004	0.189 ± 0.004
14	0.6665 ± 0.0170	3.46 ± 0.06	0.325 ± 0.005	0.345 ± 0.008	0.162 ± 0.003	0.106 ± 0.001
15	0.0327 ± 0.0005	4.76 ± 0.08	0.274 ± 0.004	0.313 ± 0.002	0.176 ± 0.001	0.144 ± 0.001
16	0.0288 ± 0.0006	3.64 ± 0.05	0.371 ± 0.009	0.378 ± 0.002	0.164 ± 0.001	0.134 ± 0.001
17	0.0298 ± 0.0005	3.25 ± 0.07	0.328 ± 0.012	0.354 ± 0.009	0.173 ± 0.005	0.117 ± 0.003
18	0.1492 ± 0.0012	3.49 ± 0.04	0.377 ± 0.009	0.407 ± 0.003	0.204 ± 0.002	0.149 ± 0.001
19	0.0376 ± 0.0009	3.50 ± 0.08	0.377 ± 0.006	0.367 ± 0.005	0.164 ± 0.004	0.131 ± 0.003
20	0.1489 ± 0.0010	3.43 ± 0.05	0.360 ± 0.012	0.386 ± 0.004	0.197 ± 0.002	0.143 ± 0.002
21	0.0287 ± 0.0011	3.55 ± 0.06	0.351 ± 0.012	0.406 ± 0.002	0.174 ± 0.001	0.144 ± 0.001

TABLE 5
Micro-minerals and toxic elements (mg/kg) present in Brazilian soluble coffee¹

Samples	Mn	Fe	Cu	Zn	Al	Sn	Cr
01	6.82 ± 0.61	39.8 ± 1.0	1.58 ± 0.01	3.17 ± 0.29	< 5.0	9.3 ± 5	< 0.10
02	5.60 ± 0.08	37.9 ± 3.6	1.38 ± 0.06	5.67 ± 0.29	< 5.0	8.5 ± 1.3	< 0.10
03	9.48 ± 0.79	32.1 ± 1.8	1.47 ± 0.01	8.50 ± 0.80	< 5.0	8.7 ± 0.5	< 0.10
04	14.24 ± 0.67	31.6 ± 2.1	2.27 ± 0.11	8.17 ± 0.29	< 5.0	9.0 ± 0.5	< 0.10
05	27.90 ± 1.00	39.9 ± 2.9	2.33 ± 0.23	10.67 ± 1.44	< 5.0	7.6 ± 1.0	< 0.10
06	12.20 ± 0.54	31.1 ± 1.7	2.15 ± 0.13	7.75 ± 0.35	< 5.0	6.4 ± 0.4	< 0.10
07	14.52 ± 1.35	144.3 ± 8.8	1.26 ± 0.15	10.00 ± 0.71	118.6 ± 6.3	8.8 ± 1.1	< 0.10
08	15.18 ± 1.58	107.8 ± 7.2	0.53 ± 0.02	15.00 ± 1.00	37.9 ± 4.1	6.4 ± 1.3	< 0.10
09	21.27 ± 0.82	287.2 ± 13.4	1.87 ± 0.14	15.17 ± 0.76	166.4 ± 5.0	9.9 ± 0.7	< 0.10
10	15.06 ± 0.60	48.9 ± 2.5	1.18 ± 0.11	7.25 ± 0.35	< 5.0	8.9 ± 0.6	< 0.10
11	8.48 ± 0.17	24.6 ± 1.7	1.09 ± 0.01	4.33 ± 0.50	< 5.0	9.4 ± 1.0	< 0.10
12	4.26 ± 0.05	17.8 ± 2.0	0.95 ± 0.09	3.80 ± 0.29	22.7 ± 0.6	9.4 ± 1.2	< 0.10
13	38.85 ± 0.58	450.8 ± 5.0	1.51 ± 0.06	7.83 ± 0.29	148.2 ± 5.4	8.5 ± 1.1	0.52 ± 0.02
14	19.38 ± 0.76	46.2 ± 0.4	0.50 ± 0.03	5.33 ± 0.29	19.7 ± 1.5	7.7 ± 0.6	< 0.10
15	22.66 ± 0.46	256.3 ± 2.0	1.72 ± 0.20	10.17 ± 0.76	233.4 ± 7.4	11.6 ± 0.9	< 0.10
16	13.86 ± 0.26	14.0 ± 0.3	0.51 ± 0.01	3.50 ± 0.3	< 5.0	9.6 ± 0.8	< 0.10
17	3.62 ± 0.08	16.8 ± 0.2	0.54 ± 0.01	3.33 ± 0.29	< 5.0	9.6 ± 0.7	< 0.10
18	18.06 ± 0.56	37.0 ± 1.5	0.87 ± 0.02	4.33 ± 0.29	24.4 ± 2.2	8.5 ± 1.1	< 0.10
19	13.38 ± 0.32	17.8 ± 1.6	0.79 ± 0.01	7.83 ± 0.29	< 5.0	8.5 ± 0.7	< 0.10
20	17.70 ± 0.62	38.2 ± 1.4	1.32 ± 0.07	4.00 ± 0.38	16.0 ± 2.2	8.1 ± 0.9	< 0.10
21	14.03 ± 0.36	47.4 ± 3.2	1.02 ± 0.06	5.17 ± 0.29	< 5.0	6.9 ± 0.9	< 0.10

¹ < not detectable at level.

The toxic elements were present in the following concentrations: Al not detected to the level of 5.0 mg/kg (57% of the samples), 16–38 mg/kg (five samples representing 24% of the total samples) and 119–233 mg/kg (four samples representing 19% of the total samples); Sn 6–12 mg/kg; Ni not detected to the level of 2.50 mg/kg; Cd not detected to the level of 0.25 mg/kg; Sb not detected to the level of 2.00 mg/kg; Pb not detected to the level of 0.45 mg/kg. Although a group of samples presented various values of Al, the maximum content defined by the Brazilian Food Legislation is 20 000 mg/kg, therefore well above that found in the different analyzed samples. Koch *et al.* (1989) reported the presence of Al in coffee from 20.7–30.6 mg/kg, being close to 16–38 mg/kg found in five samples. The presence of Sn (6–12 mg/kg) is below the limit specified by the Brazilian Food Legislation: maximum content of 150 mg/kg. The values defined by the Brazilian Food Legislation for the other elements are Ni 5.00 mg/kg maximum; Sb 2.00 mg/kg maximum; Cd 1.00 mg/kg maximum; Cr 0.10 mg/kg maximum and Pb 1.00 mg/kg maximum. Ni, Sb, Cd, Cr and Pb were not detected to the levels of concentration defined by the Brazilian Food Legislation except for Cr in sample 13 (0.52 ± 0.02 mg/kg), indicating contamination of the product. There is a great probability that this contamination source is attributed to the packing used by the manufacturer, that is, the metal container.

To identify the relationship among the analyzed samples, the technique of pattern recognition based on Principal Components Analysis and Agglomerative Hierarchical Clustering was performed using the Statistical Package for Social Sciences (SPSS) (Norusis, 1993; Bruns and Faigle, 1985; Adams, 1995; Kaufmann, 1997; Penninckx *et al.*, 1996). Figure 2 shows the Principal Components Analysis axes in two dimensions obtained by evaluation of the concentrations of elements Na, K, Mg, Al, P, S, Ca, Mn, Fe, Sn, Cu and Zn in the samples of soluble coffee studied and Fig. 3 shows the Agglomerative Hierarchical Clustering.

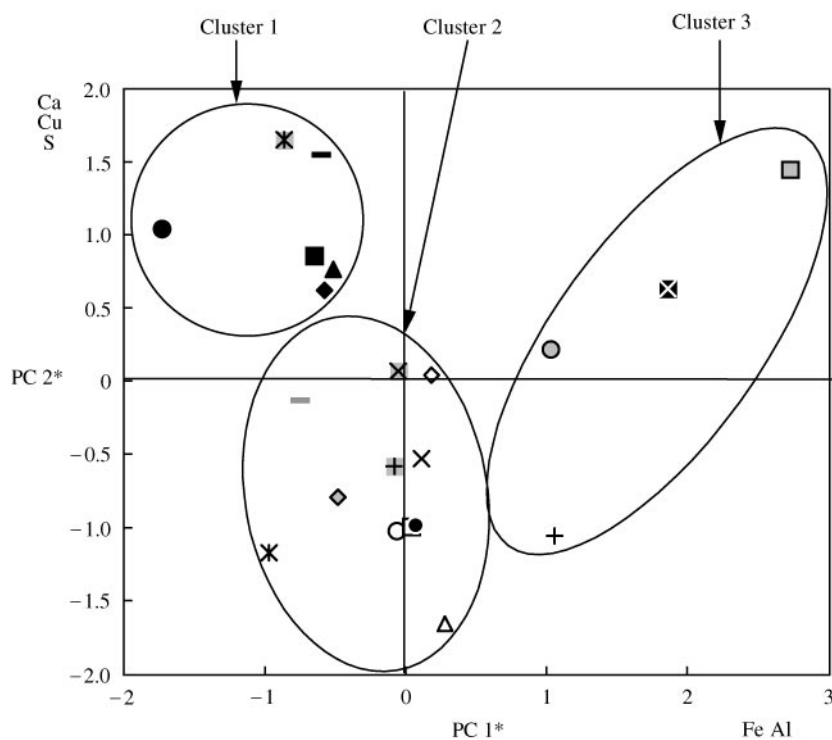


FIGURE 2. Similarity among the studied coffee samples: $\diamond$, sample 01; $\times$, sample 02; $\otimes$, sample 03; $\blacksquare$, sample 04; $\otimes$, sample 05; $\bullet$, sample 06; $+$, sample 07; $\times$, sample 08; $\odot$, sample 09; $\blacklozenge$, sample 10; $\circ$, sample 11; $\square$, sample 12; $\blacksquare$, sample 13; $\triangle$, sample 14; $\otimes$, sample 15; $\oplus$, sample 16; $\blacksquare$, sample 17; $\blacksquare$, sample 18; $\blacklozenge$, sample 19; $\blacktriangle$, sample 20; $\blacksquare$, sample 21. *Principal components.

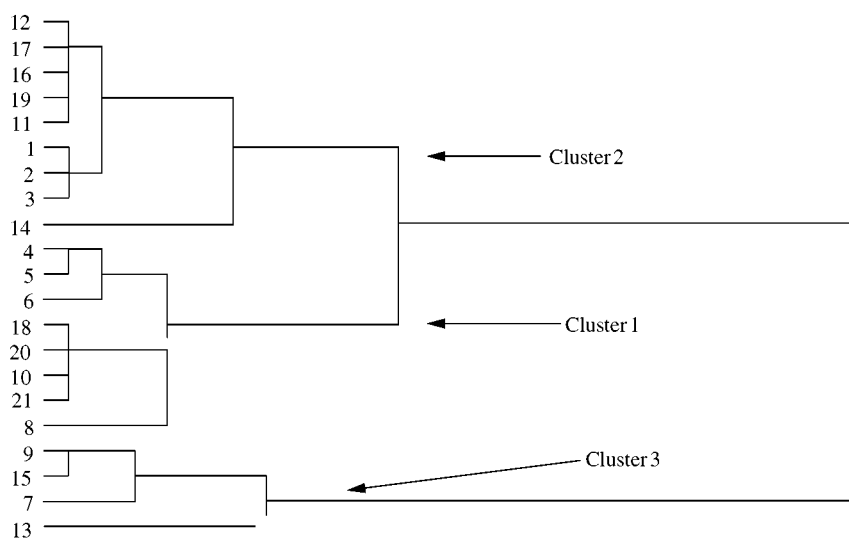


FIGURE 3. Agglomerative Hierarchical Clustering Analysis.

Through the Principal Components Analysis (Fig. 2), it is observed that the majority of the samples have different mineral compositions resulting in a dispersion in the plan defined by the components 1 and 2 (PC 1 and PC 2). However, it is possible to observe three clusters with similarities in the mineral compositions: in cluster 1 the samples have positive correlation with component 1 (PC 1) and present significant concentrations of the elements Ca, Cu and S. Cluster 2 involves the largest number of samples representing the products that are sold more in the Brazilian market and also in the states (Paraná, São Paulo and Minas Gerais); in this cluster, sample 14 presents less similarity due to its high content of Na. Cluster 3 presents three samples (13, 9 and 15) with positive correlation with the two components and sample 7 has positive correlation with component 2. These samples are rich in nutrients and present high amounts of Fe and Al. Sample 13 has high positive correlation with the two components and showed a high value for Cr. This is the only sample from the state of Espírito Santo, which has different conditions of soil where the coffee is cultivated and/or different industrial processes for the production of soluble coffee and storing (in metal containers). The results of the Principal Components Analysis are confirmed by the Agglomerative Hierarchical Clustering. The method of *Ward* (Norusis, 1993; Bruns and Faigle, 1985; Adams, 1995; Kaufmann, 1997; Penninckx *et al.*, 1996) was applied to form the clusters presented in the dendrogram shown in Fig. 3. In the three clusters it was observed that all the samples of the same mark (Cluster 1: samples 4, 5, 18 and 20; Cluster 2: samples 11, 16 and 19; samples 12 and 17; samples 1, 2 and 3; Cluster 3: samples 9 and 15) presented similar mineral compositions.

CONCLUSIONS

The samples of soluble coffee from the southeast Brazilian market present differences in the content of the elements studied, but the application of Principal Component Analysis and the Agglomerative Hierarchical Clustering can easily identify the samples with different mineral compositions, clustering most similar samples and the chemical elements contributing greatly to these differences. These variations of mineral compositions among the samples of soluble coffee can indicate differences in the process of industrial production and/or factors that influence the cultivation of the coffee plant, such as the type of soil, the use of fertilizers with different chemical compositions and the ambient conditions. However, the values determined were acceptable for human consumption considering both nutritional and toxic aspects, with the exception of one sample presenting a Cr value above that recommended by the Brazilian Food Legislation, and whose contamination can be attributed to its metal container.

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